



Welcome to Financial Accounting

Tuck Business Bridge Program

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Day 5 Agenda

- **Accounting Equation – Revisited**
- **Financial Statements – Revisited**
- **Three Statement Model - Revisited**
 - Historical Financial Statement Data
- **Financial Statement Analysis**
- **Financial Statement Forecasting**
 - Assumptions
 - Sensitivity Analysis
 - Free Cash Flows
- **Class Recap**
- **Planning Ahead – Capstone Project Accounting Lab Tomorrow**



Accounting Equation

Draft out Expanded Accounting Equation



Financial Statements

Draft out Financial Statements

Income Statement

Balance Sheet

Statement of Retained Earnings

Statement of Cash Flows

*****show articulation*****



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3-statement model

Enter historical data, as applicable

Balance Sheet

Income Statement

Statement of Cash Flows

Analyze Financial Statements

Forecast Financial Statements

Forecast Free Cash Flows



Historical Data

3 statement model template



Historical Data

Download 3 years of Financial Statements

Balance Sheet

Income Statement

Statement of Cash Flows

Using 3 statement model template, enter financial statement data, as applicable

Balance Sheet

Income Statement

Statement of Cash Flows

Remember balance sheet must balance



Financial Statement Analysis



Financial Statement Analysis

Historical

Horizontal Analysis

Vertical Analysis

Ratio Analysis



Financial Statement Analysis

Historical

Income Statement Analysis Section



Financial Statement Analysis

Historical

Horizontal Analysis

Revenue Growth

$\text{Change in Revenue} / \text{Revenue in prior period}$



Financial Statement Analysis

Historical

Vertical Analysis

Costs

Gross Profit Margin

Gross Profit / Revenue



Financial Statement Analysis

Historical

Vertical Analysis

SG&A Expense (Percent of Revenue)

$\text{SG\&A Expense} / \text{Revenue}$



Financial Statement Analysis

Historical

Ratio Analysis

Depreciation (Percent of Cap Ex)

capital expenditures

Depreciation Expense / Cap Ex



Financial Statement Analysis

Historical

Balance Sheet Analysis Section



Financial Statement Analysis

Historical Ratio Analysis

Working Capital

Days or %

AR Days (Revenue basis)

$AR/Revenue \times 365$

Inventory Days (COGS basis)

$Inventory/COGS \times 365$

Other Current Assets (Percentage of Revenue)

$Other\ Current\ Assets/Revenue$

Accounts Payable Days (COGS basis)

$AP/COGS \times 365$

Accrued Liabilities (Percent of SG&A)

$Accrued\ Liabilities/SG\&A$

Other Current Liabilities (Percentage of SG&A)

$Other\ Current\ Liabilities/SG\&A$



Financial Statement Analysis

Historical

Net Working Capital

Add AR

Add Inventory

Add Other Current Assets

Less AP

Less Accrued Liabilities

Less Other Current Liabilities



Financial Statement Analysis

Historical

Ratio Analysis

Capital Expenditures

%

Capital Expenditures (Percent of Revenue)

Capital Expenditures/Revenue



Financial Statement Analysis

Historical

Consider what the numbers may be telling you.

Consider the trend.

Understand the limitations.



Financial Statement Analysis

Historical

Limitations

Just one part of the “story”

Based on historical data

Based on specific time period



Financial Statement Analysis

Historical

Benchmark

Consider competitor / comparison data

Consider industry data

Realize the limitations with benchmarking

Supplement quantitative analysis with qualitative analysis



Financial Statement Analysis

Historical

Income Statement Analysis Section Completed

Balance Sheet Analysis Section Completed

Next steps...



Financial Statement Analysis

Forecast



Financial Statement Analysis

Forecast

Consider forecast assumptions and value drivers.



Financial Statement Analysis

Forecast

Income Statement Analysis Section



Financial Statement Analysis

Forecast

Consider Revenue Growth assumptions



Financial Statement Analysis

Assumptions

Revenue Growth Assumptions

Change in Revenue / Revenue in prior period

Baseline



Financial Statement Analysis

Assumptions

Sensitivity Analysis

Revenue Growth Assumptions

Change in Revenue / Revenue in prior period

Optimistic

Baseline

Pessimistic



Financial Statement Analysis

Forecast

Consider Cost assumptions



Financial Statement Analysis

Assumptions

Cost Assumptions

$$\text{Gross Profit Margin} = \text{Gross profit/revenue}$$

Baseline

$$(\text{Gross Profit is Revenue} - \text{COGS})$$



Financial Statement Analysis

Assumptions

Sensitivity Analysis

Cost Assumptions

Gross Profit Margin = Gross profit/revenue

Optimistic

Baseline

Pessimistic

(Gross Profit is Revenue – COGS)



Financial Statement Analysis

Assumptions

Vertical Analysis

SG&A Expense (Percent of Revenue)

SG&A Expense / Revenue



Financial Statement Analysis

Assumptions

Ratio Analysis

Depreciation (Percent of Cap Ex)

capital expenditures

Depreciation Expense / Cap Ex



Financial Statement Analysis

Forecast

Balance Sheet Analysis Section



Financial Statement Analysis

Forecast

Consider Investment assumptions



Financial Statement Analysis

Assumptions

Investments: Working Capital

Working Capital

Days or %

AR Days (Revenue basis)

$AR/Revenue \times 365$

Inventory Days (COGS basis)

$Inventory/COGS \times 365$

Other Current Assets (Percentage of Revenue)

$Other\ Current\ Assets/Revenue$

Accounts Payable Days (COGS basis)

$AP/COGS \times 365$

Accrued Liabilities (Percent of SG&A)

$Accrued\ Liabilities/SG\&A$

Other Current Liabilities (Percentage of SG&A)

$Other\ Current\ Liabilities/SG\&A$



Financial Statement Analysis

Based on your assumptions:

Net Working Capital

Add AR

Add Inventory

Add Other Current Assets

Less AP

Less Accrued Liabilities

Less Other Current Liabilities



Financial Statement Analysis

Assumptions

Investments: Capital Expenditures (CapEx)

Capital Expenditures

%

Capital Expenditures (Percent of Revenue)

Capital Expenditures/Revenue



Financial Statement Analysis

Assumptions

Other Accounts, as needed

See notes in 3-statement model such as:

“Set at last known historical value and keep it constant.”



Supporting Schedules



Supporting Schedule

Net Property, Plant and Equipment (PPE)

Beginning of Period

Add Capital Expenditures

Less Depreciation Expense

End of Period



Supporting Schedule

Retained Earnings

Beginning of Period

Add Net Income

Less Dividends, if any

Ending of Period



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Free Cash Flows



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Free Cash Flows

Operating Income * (1-Tax Rate)

Add Depreciation Expense

Less Capital Expenditures

Less Change in Net Working Capital

Unlevered Free Cash Flows (FCF)



EBIT and EBITDA

**EBIT –
earnings before interest and taxes**

**EBITDA –
earnings before interest, taxes, depreciation and amortization**



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Review of 3-Statement Model



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Class Recap – topics included:

Accounting Concepts and Theory

Accounting Equation

Expanded Accounting Equation

Debits and Credits

Journal Entries

Accrual Accounting / Adjusting Entries

Closing Entries

Financial Statements

Balance Sheet

Income Statement

Statement of Cash Flows

Statement of Retained Earnings

Financial Statement Analysis

Financial Statement Forecasting

This Afternoon – Capstone Project Accounting Lab



Capstone Project Accounting Lab

Historical Financial Statement Data

Assumptions

Forecast Financial Statements

Forecast Free Cash Flows



Questions?

Thank you for a great week of Financial Accounting!

I look forward to this afternoon's lab and then your final capstone presentations.

